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Long-term follow-up of Ru-106/Rh-106 brachytherapy for posterior uveal melanoma

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Abstract ● **Background:** Currently available information about survival, local tumor control rates, secondary enucleation rates, and visual acuity following Ru-106/Rh-106 applicator therapy for choroidal and ciliochoroidal melanomas is limited in terms of duration of follow-up among surviving patients. ● **Methods:** The authors performed a retrospective descriptive study of the rates of survival, local treatment failure, secondary enucleation, and visual acuity decrease in 140 patients (141 eyes) with choroidal or ciliochoroidal melanoma treated by Ru-106/Rh-106 applicator radiotherapy between 1964 and 1976. Median follow-up duration among surviving patients in this series was 17.3 years (mean 18.6 years). The Kaplan-Meier method was used to estimate cumulative survival rates and event rate curves. Multivariate Cox proportional hazards modeling was used to identify prognostic clinical variables associated with the various evaluated outcomes.

● **Results:** The 15-year survival rate based on all causes of death was 48.0% (standard error=4.4%), and that based on confirmed and suspect-

ed metastatic uveal melanoma was 66.7% (standard error=4.5%). The cumulative 15-year rates of local treatment failure and secondary enucleation were 36.8% (standard error=4.4%) and 34.4% (standard error=4.5%) respectively. The cumulative 10-year rates of visual acuity loss to less than 20/200 and no light perception were 62.8% (standard error=4.4%) and 40.6% (standard error=5.2%) respectively. Prognostic factors associated with death from confirmed and suspected metastatic melanoma were greater tumor diameter and anterior tumor location. Greater tumor diameter was associated with local treatment failure. Baseline visual acuity equal to or worse than 20/200 was associated with profound visual acuity loss.

● **Conclusions:** Although a high proportion of treated eyes eventually lost a great deal of vision, and although many treated eyes ultimately underwent secondary enucleation, a substantial number of patients treated by plaque radiotherapy in this series survived for well over 10 years and retained the tumor-containing eye with a visual decrease of varying severity.

Introduction

Brachytherapy with Ru-106/Rh-106 applicators was introduced for the treatment of posterior uveal melanomas in 1964 by Lommatzsch and Vollmar [16]. Many descriptive studies summarizing the survival of treated pa-

tients, local tumor control rates, secondary enucleation rates, and changes in visual acuity following Ru-106/Rh-106 plaque therapy have since been published. The most recent comprehensive follow-up report based on our personal experience with Ru-106/Rh-106 applicator therapy for choroidal and ciliochoroidal melanomas summarized

our results in a mean postirradiation follow-up time among surviving patients of 6.7 years [15]. This paper reports the results after longer follow-up.

Patients and methods

In a retrospective, descriptive study on the first 140 patients (141 eyes) with a choroidal or ciliochoroidal melanoma treated by Ru-106/Rh-106 applicator radiotherapy in the authors' practice between 1964 and 1976, the closing date for data analysis was June 1998. All of the patients had thus been treated at least 22 years earlier. The diagnosis of choroidal or ciliochoroidal melanoma was based on the findings of comprehensive ophthalmic examination supplemented, when possible, by fluorescein angiography, ultrasonography, P-32 test, and other studies. Multiple clinical variables were evaluated preoperatively, including age (years) and sex of the patient, thickness (mm) and largest basal diameter (mm) of the tumor, location of the tumor relative to the optic disc and fovea and relative to the equator and ora serrata, and best corrected distance visual acuity in the affected eye.

All tumors were treated with a commercially available Ru-106/Rh-106 applicator which was designed to deliver a radiation dose of approximately 100 Gy to the tumor's apex during a period of 4–7 days. If a tumor showed limited or no clinical regression, a second plaque application was frequently recommended. For tumors less than 3 mm from the optic disc, the fovea, or both, supplementary xenon arc photocoagulation was occasionally recommended following plaque therapy.

Following treatment, patients were scheduled for reevaluation every 3 months for the first year, every 6 months for the second year, and at least once a year thereafter. Follow-up evaluations included measurement of best corrected distance visual acuity and ophthalmoscopic assessment of local tumor control status (complete regression, partial regression, or progression). Patients did not undergo any routine systemic screening tests for uveal melanoma metastasis as part of their standard follow-up; however, patients who developed abdominal pain or swelling, nausea, substantial weight loss, jaundice, subcutaneous nodules, etc. were evaluated at that time for the presence of metastatic disease.

Patients were followed, if possible, until death or most recent scheduled examination. If a patient had died, the cause of death was sought from review of medical records related to the terminal event. If a patient was unable to return to us for reevaluation, an attempt was made to obtain periodic follow-up by telephone or mail contact with the patient, a family member, or a local physician involved in the patient's care. A patient was considered lost to follow-up if we were unable to obtain any follow-up information. Censoring occurred at the last known contact with the patient. This reflects the duration of follow-up for each single case. Outcomes evaluated in this study included survival status (coded as dead of metastatic uveal melanoma, dead of uncertain cause, dead of non-melanoma-related cause, or alive), local tumor control status (complete regression, partial regression, insufficient regression, progression, or local relapse), secondary enucleation (no or yes), and visual acuity in the treated eye (equal to or better than 20/200 versus worse than 20/200 and equal to or better than light perception versus no light perception).

Survival rates based on deaths of any cause, deaths from metastatic uveal melanoma and non-tumor-related deaths, and event rates of local tumor failure, secondary enucleation, and decrease in visual acuity to the specified levels were estimated using the Kaplan-Meier product limit method [3]. When evaluating causes of death, any patient who had metastatic melanoma at the time of death was considered to have died of melanoma, even if the recorded immediate cause of death was another illness or condition. In addition, patients who died of unknown cause were assumed to

have died of metastatic uveal melanoma for the purposes of this study. When evaluating local tumor control status, an eye that relapsed and underwent supplementary treatment due to insufficient tumor regression or progression was regarded as a local treatment failure at that time, regardless of whether supplementary treatment subsequently yielded satisfactory local control. When evaluating visual acuity, eyes that underwent enucleation following plaque radiotherapy but still had visual acuity of light perception or better at the preenucleation examination were coded as having no light perception (NLP) on the date the enucleation was performed. For eyes that experienced a pronounced decline in visual acuity following Ru-106/Rh-106 applicator therapy, patient records were reviewed to identify (if possible) the principal mechanism responsible.

Baseline prognostic factors for death, local treatment failure, and decrease in visual acuity to NLP were evaluated using multivariate Cox proportional hazards modeling [3]. This modeling was performed as a stepdown procedure, with nominal *P* values of 0.05 to enter and 0.17 to remove.

Results

Our study group consisted of 72 men (51.4%) and 68 women (48.6%). The 140 patients ranged in age from 14 years to 82 years (mean 54.8 years) at the time of treatment. Visual acuity at baseline examination ranged from 20/14 to 1/50 (median 20/40) in the 141 study eyes. The 141 tumors ranged from 1.0 to 7.0 mm in thickness (mean 2.7 mm) and from 3.0 to 12.0 mm in largest basal diameter (mean 6.4 mm). The tumors were classified according to the TNM system [11] as T1 in 109 eyes (77.3%), T2 in 26 eyes (18.4%), and T3 in 6 eyes (4.3%). According to the COMS system [6] the tumors were small in 111 eyes (78.7%) and medium in 30 eyes (21.3%). The center of the tumor lay posterior to the equator in 12 eyes (8.5%), between the equator and ora serrata in 91 eyes (64.5%), and anterior to the ora serrata in the ciliary body in 38 eyes (27.0%).

The diagnosis of all melanomas was confirmed by a positive P-32 uptake of more than 40% [17].

Supplementary xenon arc photocoagulation was performed shortly after Ru-106/Rh-106 applicator therapy in 24 eyes (17.0%) with a tumor that extended posteriorly to less than 3 mm from the disc, the fovea, or both.

Seventeen eyes (12.1%) were retreated with brachytherapy subsequent to initial therapy because of limited tumor regression, tumor progression, or relapse following initial treatment. The interval between the first and second radiotherapy treatments ranged from 3 months to 251 months (mean 43.5 months, median 9.0 months).

Five of these eyes received both supplementary xenon arc coagulation and second plaque application.

As of the closing date of the study, 79 (56.4%) of the 140 patients were dead. Of the 79 deaths, 34 (43.0%) were attributed to metastatic uveal melanoma, 9 (11.4%) to unknown causes, and 36 (45.6%) to a cause other than metastatic melanoma. For the purposes of this study, patients dying of confirmed metastatic melanoma and those

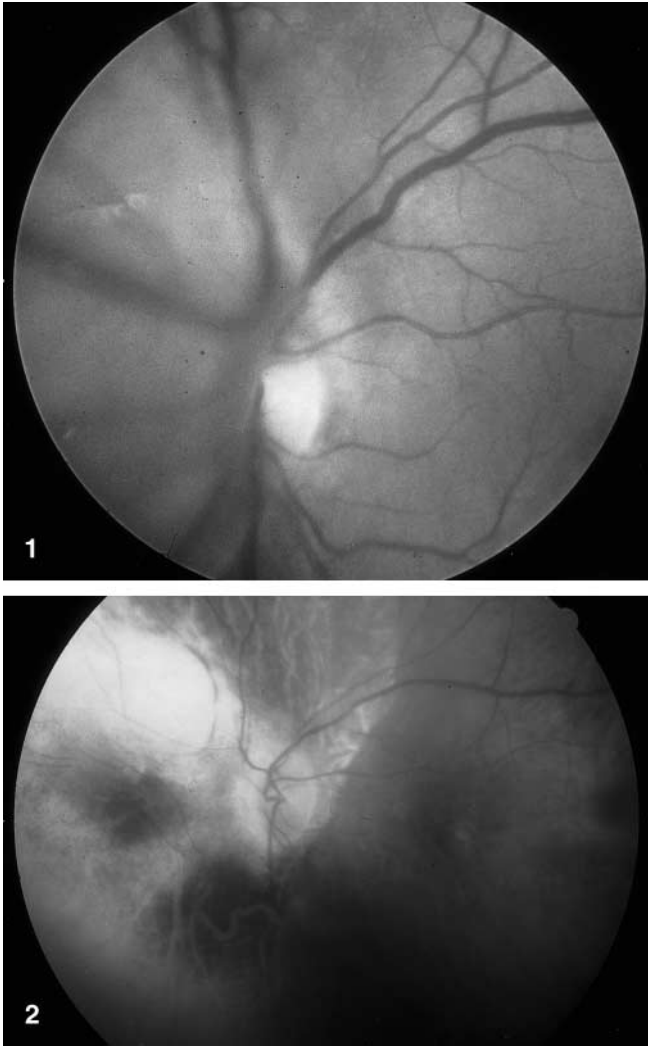


Fig. 1 Choroidal melanoma of the left eye in a 62-year-old woman. Visual acuity = 20/20

Fig. 2 Twenty years after brachytherapy with Ru-106/Rh-106. Visual acuity = 20/200 due to maculopathy

dying of unknown cause were combined under the term “confirmed and suspected metastatic melanoma” (43/79). The total duration of follow-up among the 61 surviving patients ranged from 4 years to 33.7 years (mean 18.6 years, median 17.3 years). Figure 3 shows the survival curves based on all causes of death, on confirmed metastatic melanoma-related deaths, on tumor-related deaths plus all deaths of unknown cause, and on non-melanoma-related deaths. The 5-, 10-, 15-, and 20-year survival rates considering confirmed and suspected tumor-related deaths were 89.6%, 81.0%, 66.7%, and 61.3%, respectively. For the purpose of this study the 141 tumors were divided into two groups with greatest basal tumor diameter less than or equal to 6 mm and

more than 6 mm. Figure 4 shows survival estimates based on tumor-related death, with a significant difference between these two groups ($P=0.0088$). The cumulative tumor-related death rates of patients with small tumors (less than or equal to 6 mm in basal diameter) ($n=86$) were 6.2% after 5 years and 13.9% after 10 years. The 5-year and 10-year tumor-related mortality rates of cases with a basal tumor diameter greater than 6 mm ($n=55$) were 16.6% and 26.7%, respectively.

During available follow-up, 49 (34.7%) of the 141 tumors exhibited complete regression to a flat, atrophic lesion and 41 (29.1%) showed partial regression without reactivation after first irradiation. In 6 cases (4.2%) we found partial regression, but only after a second plaque therapy. Twenty tumors (14.2%) showed limited tumor regression or no appreciable change in size following radiotherapy, which ultimately led to second brachytherapy or enucleation. Unfortunately, 17 tumors (12.1%) enlarged progressively in spite of radiotherapy, and 7 others (5.0%) relapsed locally after a variable period of partial regression. Figure 5 shows the cumulative event rate curve for local treatment failure (35.5%) of in this study group. The 15-year event rate for local treatment failure was 36.8% (standard error=4.4%). One case was not assessable.

Secondary enucleation was performed on 44 (31.2%) of the 141 study eyes. Out of these 44 enucleated eyes, 7 tumors showed new growth after irradiation, 16 enlarged progressively and in 12 eyes the tumor exhibited “failure to shrink”. Eight tumors received a second plaque therapy before enucleation. The main clinical indications for enucleation of the 44 eyes were as follows: minimal if any clinical regression (“failure to shrink”) in 7, progressive enlargement following plaque radiotherapy in 15, delayed local relapse in 7, and radiation-related ocular complications leading to a blind, painful, or unsightly eye in 15 eyes. Figure 6 shows the cumulative event rate curve for secondary enucleation in this series. The 15-year event rate for secondary enucleation was 34.4% (standard error=4.5%). A comparison between the cumulative survival rates (based on tumor-related death) of enucleated and non-enucleated patients after brachytherapy was performed. There was no statistically significant difference between these two survival curves as shown in Fig. 7 (log rank test: $P=0.9369$).

During follow-up, visual acuity decreased to less than 20/200 in 95 eyes (67.4%) and to NLP in 48 eyes (34.0%). Figure 8 shows the cumulative event rate curves for visual decline to these two specified levels. The most frequently identified mechanism for visual loss in these patients included radiation retinopathy involving the macula, radiation-induced optic neuropathy, exudative retinal detachment, vitreous hemorrhage, radiation-induced cataract, and enucleation.

The results of the analysis of prognostic factors using Cox proportional hazards modeling are summarized in

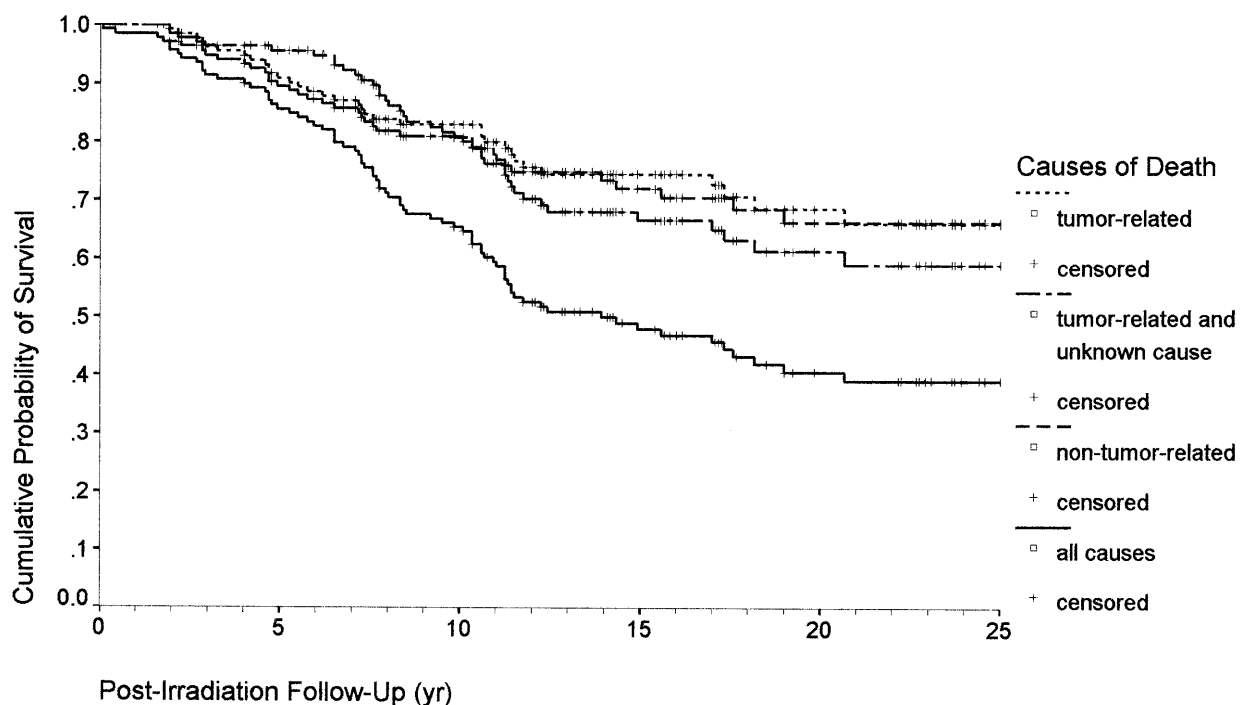


Fig. 3 Kaplan-Meier cumulative survival curves of 140 patients with choroidal or ciliochoroidal melanoma treated by Ru-106/Rh-106 applicator between 1964 and 1976: deaths of any cause, meta-

static melanoma-related deaths, confirmed plus suspected melanoma-related deaths, and non-tumor-related deaths

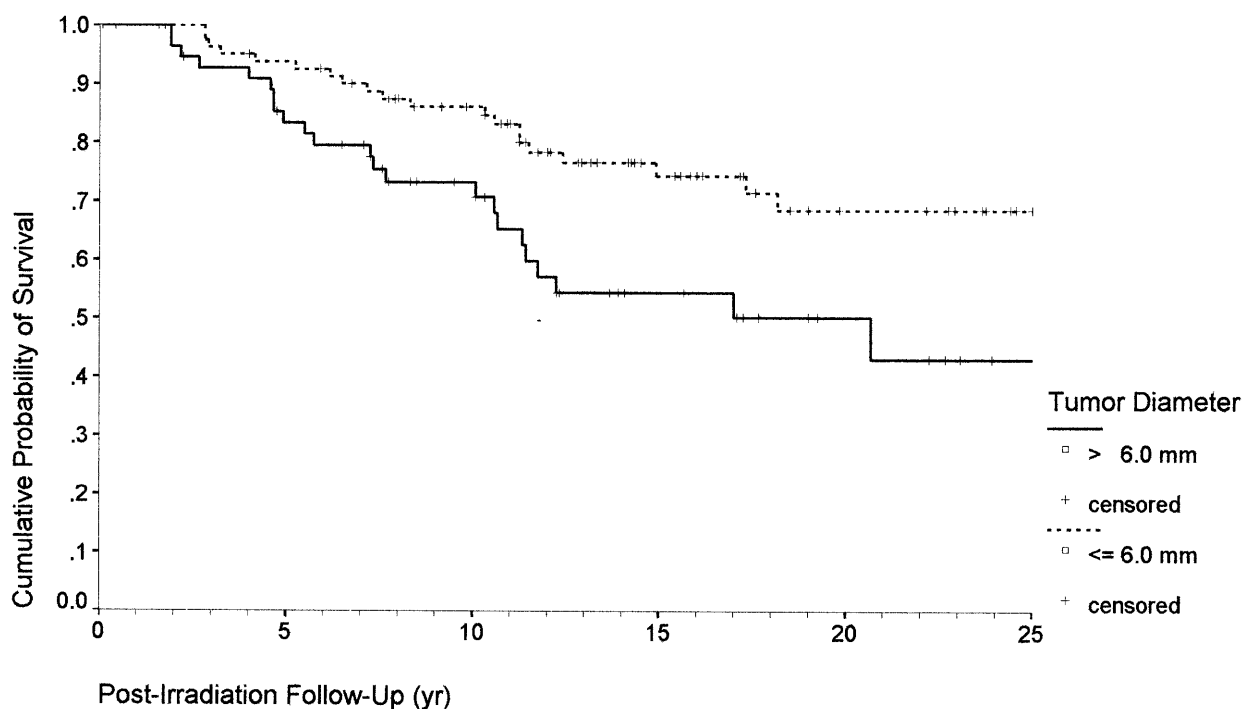


Fig. 4 Kaplan-Meier cumulative survival estimates of the patients with greatest tumor diameter equal to or less than 6.0 mm versus those with a tumor of diameter greater than 6.0 mm (based on tu-

mor-related death). The difference between the two curves was statistically significant (log rank-test $P=0.0088$)

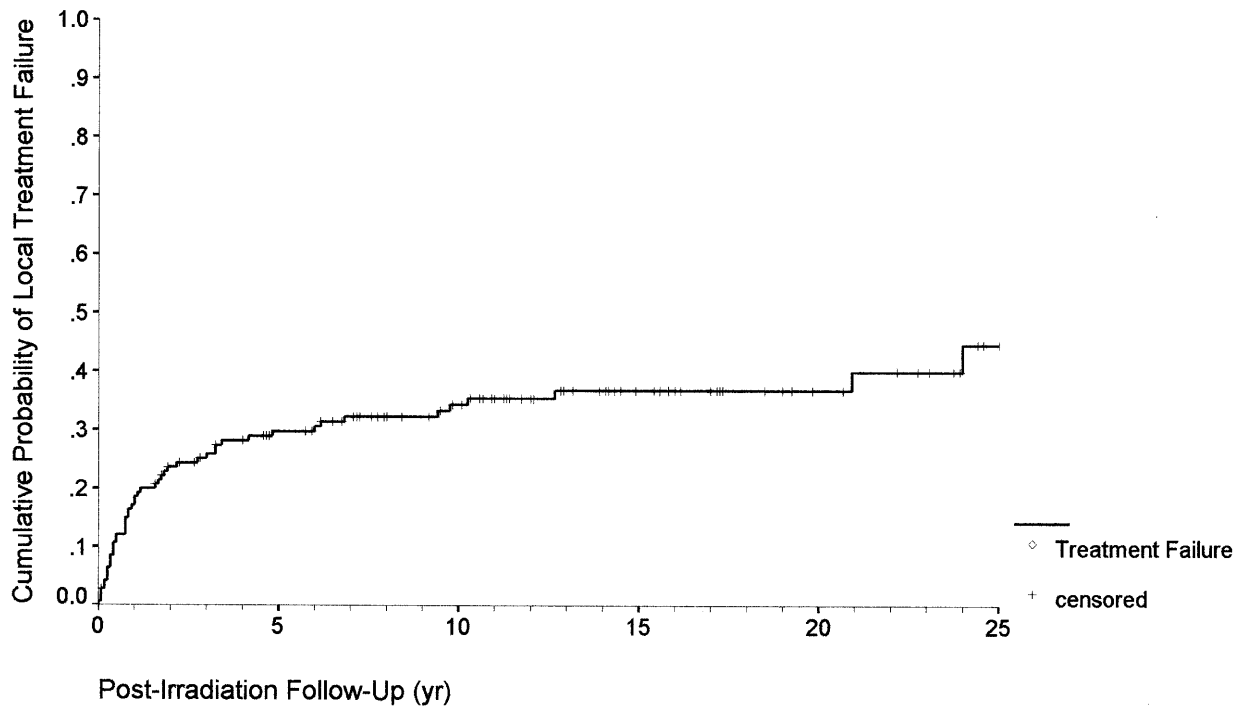


Fig. 5 Kaplan-Meier event rate curve for local treatment failure following Ru-106/Rh-106 plaque therapy in 141 eyes with choroidal or ciliochoroidal melanoma

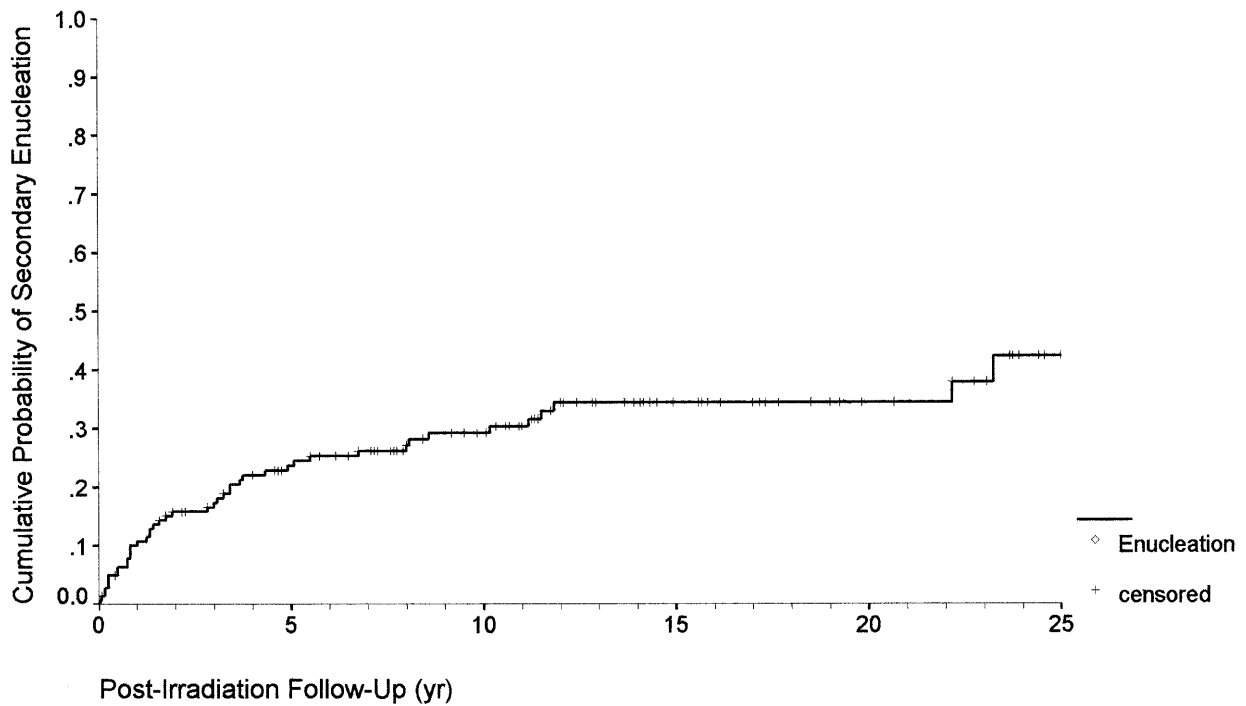


Fig. 6 Kaplan-Meier event rate curve for secondary enucleation following Ru-106/Rh-106 plaque therapy of 141 eyes with choroidal or ciliochoroidal melanoma

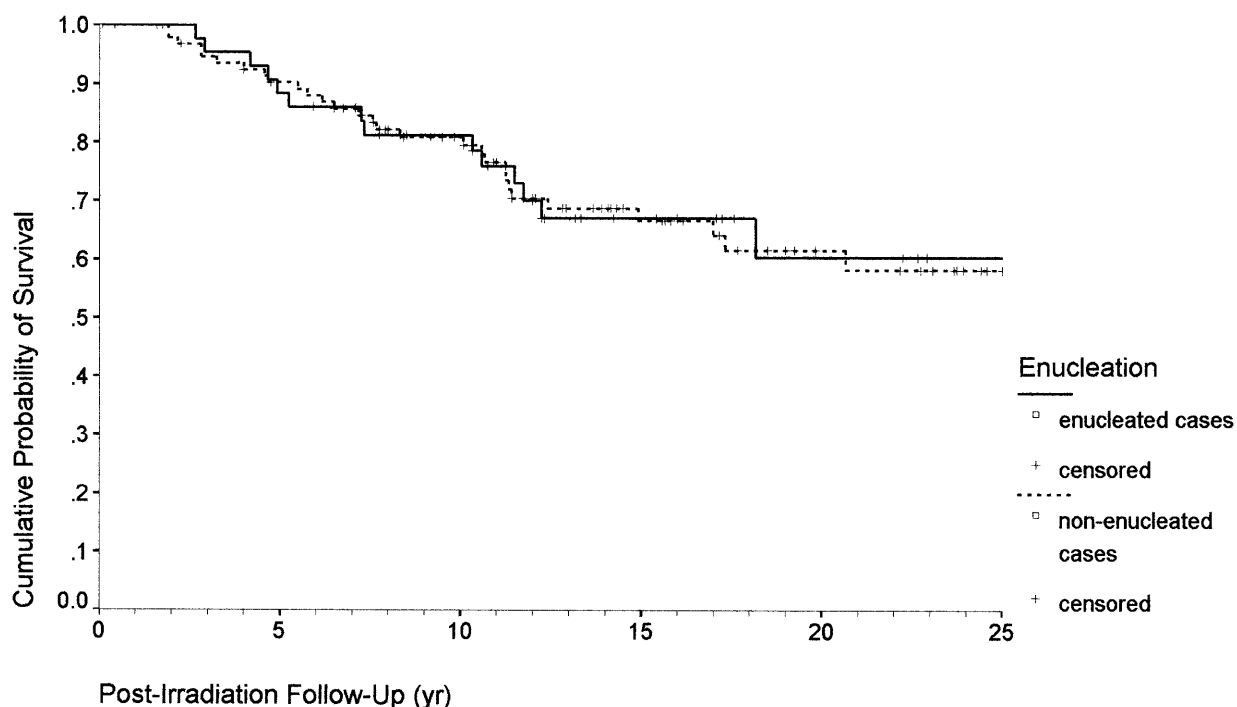


Fig. 7 Kaplan-Meier cumulative survival curves of 140 patients with choroidal and ciliochoroidal melanoma (based on tumor-related death). The *upper curve* includes patients with enucleation

after brachytherapy, while the *lower curve* summarizes the patients without secondary enucleation. The difference between the two curves is not statistically significant (log rank-test $P=0.9369$)

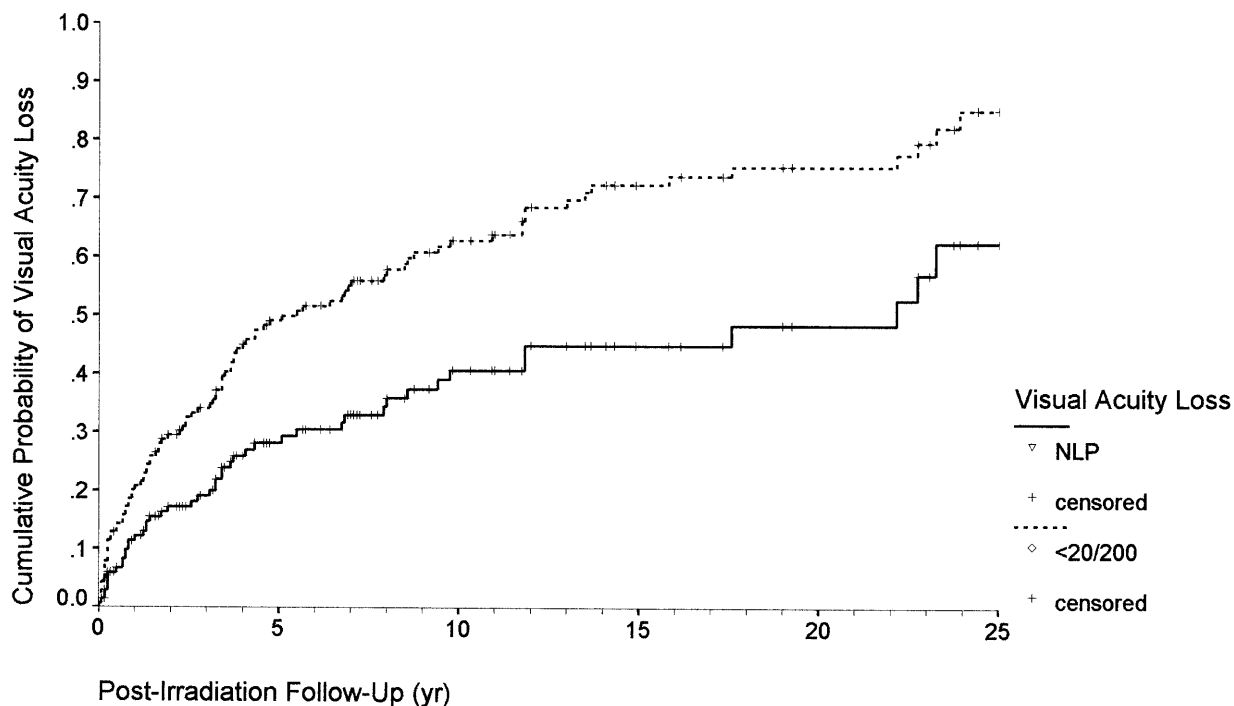


Fig. 8 Kaplan-Meier event rate curves for visual acuity decrease to less than 20/200 (*upper curve*) and to no light perception (NLP; *lower curve*) following Ru-106/Rh-106 applicator therapy in 141 eyes with choroidal or ciliochoroidal melanoma. All eyes that

were enucleated secondarily but had visual acuity of light perception or better at examination just prior to enucleation were regarded as having no light perception at the date of enucleation

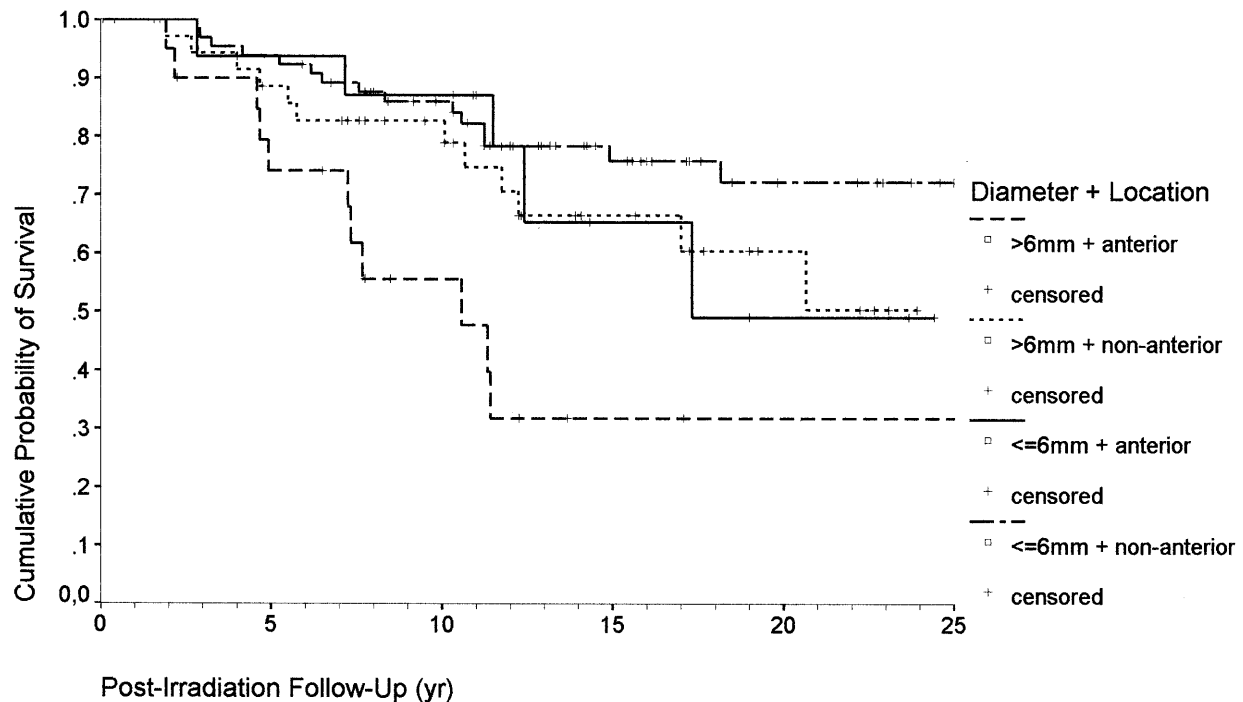


Fig. 9 Kaplan-Meier cumulative survival curves of 140 patients considering melanoma-related deaths, by largest tumor diameter (≤ 6 mm / > 6 mm) and tumor location (anterior / non-anterior)

Table 1 Best multivariate Cox regression for predicting death from metastatic melanoma, treatment failure, and visual acuity loss in 140 patients with posterior uveal melanoma

Dependent variable	Independent covariates of program selected model	Relative risk	95% Confidence interval	P-value
Confirmed and suspected tumor-related death	Larger tumor diameter (> 6 mm)	2.1	1.1–3.9	<u>0.0176</u>
	Anterior tumor location	1.9	1.0–3.5	0.0573
Treatment failure	Larger tumor diameter (> 6 mm)	2.0	1.1–3.5	<u>0.0182</u>
	Age (> 55 years)	0.6	0.4–1.1	0.1274
Visual acuity loss to no light perception	Baseline visual acuity $\leq 20/200$	2.8	1.3–5.7	<u>0.0054</u>

Table 1. The analysis of prognostic factors using deaths from confirmed and suspected metastatic melanoma as the endpoint identified anterior tumor location and larger tumor diameter as independent prognostic covariates. The relative risk of death from tumor-related causes was 2.1 times higher in patients with larger tumor diameter than in those with smaller tumors. The patients with anterior tumor location had a 1.9 times higher relative risk of death from melanoma metastasis than those with posteriorly located tumors. Figure 9 shows the cumulative survival curves based on greatest tumor diameter and tumor location. Patients with tumor diameter greater than 6 mm and anterior tumor location had a significantly worse survival rate than those with diameter greater than 6 mm and non-anterior location ($P=0.0003$). Cases with tumor diameter greater than 6 mm and anterior location

had a significantly worse survival rate than those with less than or equal to 6 mm in diameter and anterior location ($P=0.0364$) or non-anterior location ($P=0.0361$). The other pairwise tests were not significant.

The analysis of prognostic factors using local treatment failure as the endpoint identified larger tumor diameter as prognostic covariate. The relative risk of treatment failure was 2.0 times higher in cases with larger tumors than in those with smaller melanomas.

The analysis of prognostic factors using secondary enucleation as the endpoint did not identify any prognostic covariates.

The analysis of prognostic factors using decrease in visual acuity to NLP as the endpoint identified baseline visual acuity $< 20/200$ as prognostic covariate. The relative risk of decline in visual acuity to NLP was 2.8 times

higher in patients with a pretreatment baseline visual acuity of worse than or equal to 20/200 than in those with visual acuity better than 20/200 at baseline.

Discussion

Irradiation methods, dosage, and therapeutic results of brachytherapy with Ru-160/Rh-106 plaques have been described in earlier papers [13–16, 20]. Most of the critical objections [18] concerned the lack of both long-term follow-up and randomized prospective studies. The randomized Collaborative Ocular Melanoma Study (COMS) [5, 6] is now under way in the USA, although this is being conducted with I-125 rather than with Ru-106/Rh-106 plaques. In the near future this study may reveal whether radiotherapy or enucleation of small and medium-sized melanomas results in a higher survival rate.

Our retrospective study presents the findings of long-term follow-up examinations of patients suffering from small and medium-sized posterior uveal melanomas. The mean greatest tumor diameter of our 141 studied eyes was 6.4 mm before treatment. One may question whether the small tumors in our study are definitely all melanomas. It is unlikely that our selected group of small (basal diameter less than or equal to 6 mm) tumors mainly consists of nevi or other benign lesions, because all tumors had a positive P-32 uptake of more than 40%. Our experience with the P-32 test has shown an accuracy rate of 95% [17]. Among the group of small tumors 26 eyes had to be enucleated. In all of them the diagnosis “melanoma” was confirmed by histological examination. In this groups of patients treated more than 20 years ago, no biopsies were performed before therapy. This may result in a false diagnosis of melanoma in a very small number of cases. We evaluated only the charts of patients whose first therapeutic procedure had been performed at least 22 years previously. The mean follow-up period among the surviving patients was 18.6 years with a maximum of 33.7 years. In 24 eyes (17.0%) with tumor close to the macula or optic disc we performed additional xenon coagulation. In 17 eyes (12.1%) a second plaque application was necessary because the first applicator was not placed properly or the tumor showed insufficient regression, progression, or relapse after the first treatment.

On account of failure of radiation treatment or radiation-related ocular complications leading to loss of function we had to enucleate 44 eyes (31.2%). This rate is higher than reported for a previous study which was based on a larger patient group but shorter follow-up [2]. In 7 eyes of our series the indication for enucleation was “failure to shrink”. In other studies, especially with charged particles, those eyes with stabilized tumors are retained because these tumors do no worse than tumors that shrink [4]. This could be the reason for the higher local failure rate and the lack of difference in the survival

of enucleated and non-enucleated patients in our series compared to other reports.

In 63.8% of all irradiated eyes the tumors showed an excellent regression pattern and left flat scars or insignificant elevated pigmented remnants of the former prominent tumor (Figs. 1, 2). This finding is comparable with the results published in previous papers [15]. Some 35.5% of irradiated tumors showed insufficient regression or no appreciable change in size, tumor progression, or relapse after first brachytherapy, leading to enucleation or second plaque therapy. The proportion of treatment failures was higher than reported in a previous study of treatment with charged particles [4].

Special attention should be paid to the fact that enucleation due to tumor regrowth or complications after irradiation reached a maximum during the first 3 years after treatment (24/44 cases). Similar observations have been reported in a study of the long-term outcome after helium ion irradiation [4]. But even after more than 20 years of follow-up we observed two cases with tumor recurrence, which finally led to enucleation. There was no difference between the survival rates of enucleated and non-enucleated patients as shown in Fig. 7 ($P=0.9369$). This finding contradicts published results for patients treated with proton irradiation [9, 10], which demonstrated that the only factor which had no detrimental effect on patient survival was an absence of living tumor cells within the enucleated eye.

A significant proportion of our patients (36/79 deaths) died from non-melanoma-related causes. The other 43 died from confirmed and suspected metastatic melanoma. For this reason it is necessary to distinguish clearly between survival rates based on melanoma and on non-melanoma-related cause of death, as shown in Fig. 3 [12]. The 5-, 10-, 15-, and 20-year cumulative survival rates based on confirmed and suspected tumor-related deaths were 89.6%, 81.0%, 66.7% and 61.3%, respectively. In comparison with published charged particles data for larger tumors we found similar survival rates after 5 and 10 years [4].

The cumulative 5-year tumor-related death rate of our patients with small tumors was 6.2% and the 5-year all-cause mortality rate was 6.9%. Other studies of small tumors treated with various forms of irradiation reported a 5-year tumor-related mortality rate of approximately 13% [8, 15]; the 5-year all-cause death rate in reported series ranged from 9% to 18% [1, 7, 19]. This contradicts other publications with small melanomas managed by observation, in which only 3% had melanoma metastasis by 5 years [21]. The COMS [5] found a tumor-related mortality rate of 1% and an all-cause death rate of 6% after 5 years. The differences between these results and our own can probably be explained by different approaches in the measurement of tumor extension (ultrasonography versus optical methods). The higher tumor-related mortality rate we reported in our study of

small tumors confirms the diagnosis "melanoma", and it seems unlikely that these tumors were benign lesions.

A disappointing finding is the slow deterioration in visual acuity as time progresses, whether or not there was adequate tumor control. The likelihood of maintaining the pretreatment visual acuity is extremely low. The reasons for this are scarification with tissue atrophy around the tumor base and radiation-induced retinopathy and optic neuropathy. In contrast to experience with proton beam irradiation, helium ions [4], and gamma rays (Co-60, I-125) there was no incidence of keratopathy due to dry eye, loss of eye lashes, or totally opaque radiation cataract, and only five cases of secondary hemorrhagic glaucoma were observed. Multiple subconjunctival pigmentations at the area in which the applicator was positioned were apparent in a small number of cases.

Histological examinations of these spots showed pigment-loaded macrophages and no sign of extrascleral tumor extension.

In conclusion, patients suffering from small and medium-sized choroidal melanomas who are treated with Ru-106/Rh-106 plaques display a high tumor control rate of 63.8% after first irradiation and a cumulative survival rate based on confirmed and suspected melanoma-related deaths of 66.7% after 15 years, but the prospects of preventing gradual deterioration of visual acuity are limited.

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